

CST8912 – Cloud Solution Architecture

Graded Lab Activity #9

Introduction: Microsoft Azure provides a comprehensive suite of tools to protect data according to an organization's security and compliance requirements. One fundamental aspect of data security is encryption at rest, which ensures that data stored in Azure services is protected against unauthorized access and physical threats. Azure SQL Database, in particular, includes robust security features such as encryption, data classification, and auditing to help safeguard sensitive data and meet regulatory standards. This lab focuses on exploring these features to enhance the security and data privacy of an Azure SQL Database.

Objective: The main objective of this lab is to understand and implement the security features offered by Microsoft Azure for an Azure SQL Database. By completing this lab, you will:

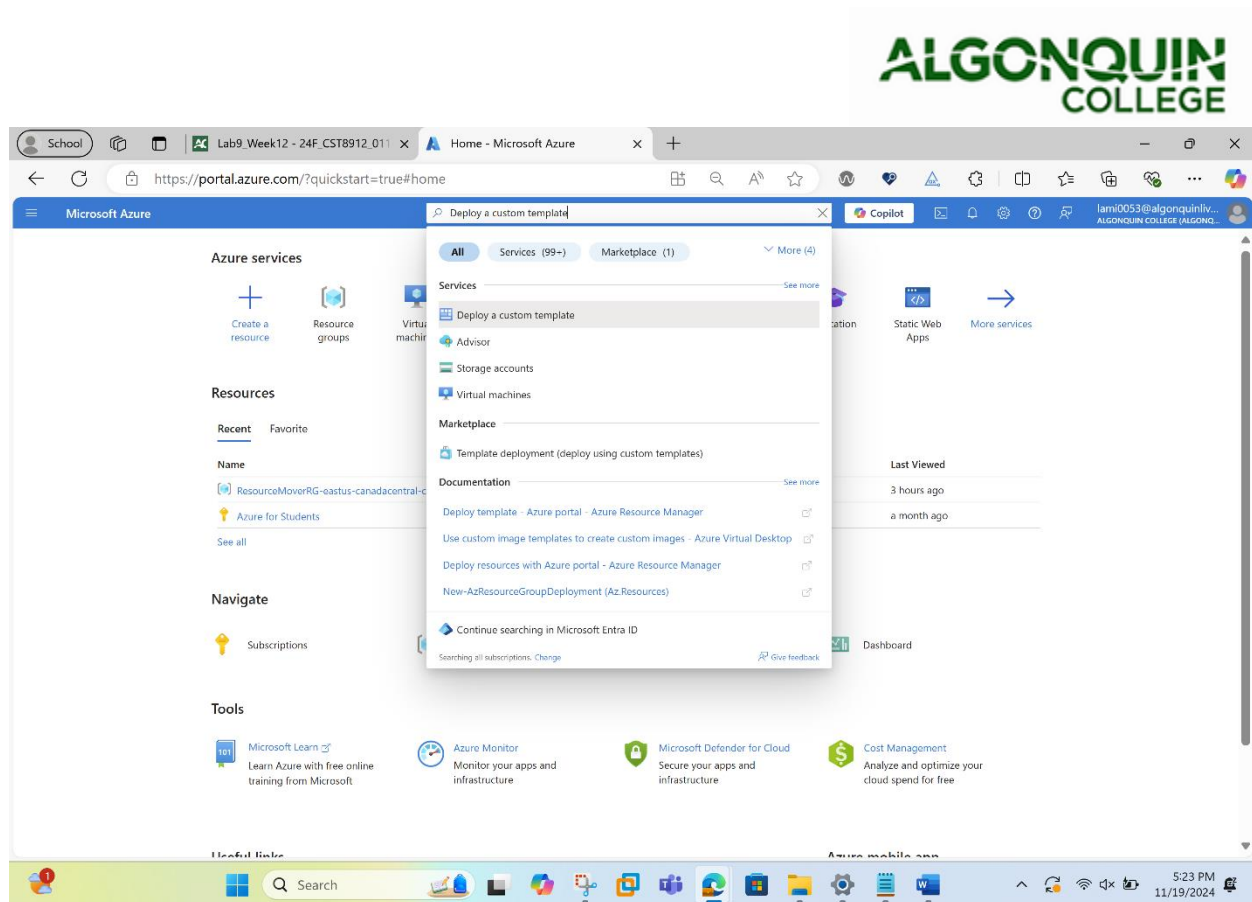
- Deploy an Azure SQL Database.
- Configure advanced data protection mechanisms to defend against SQL injection and data exfiltration.
- Implement data classification to identify and organize data into categories, such as Confidential.
- Set up auditing to monitor database activities and log events for security and compliance.
- Clean up resources after completing the lab.

Instructions:

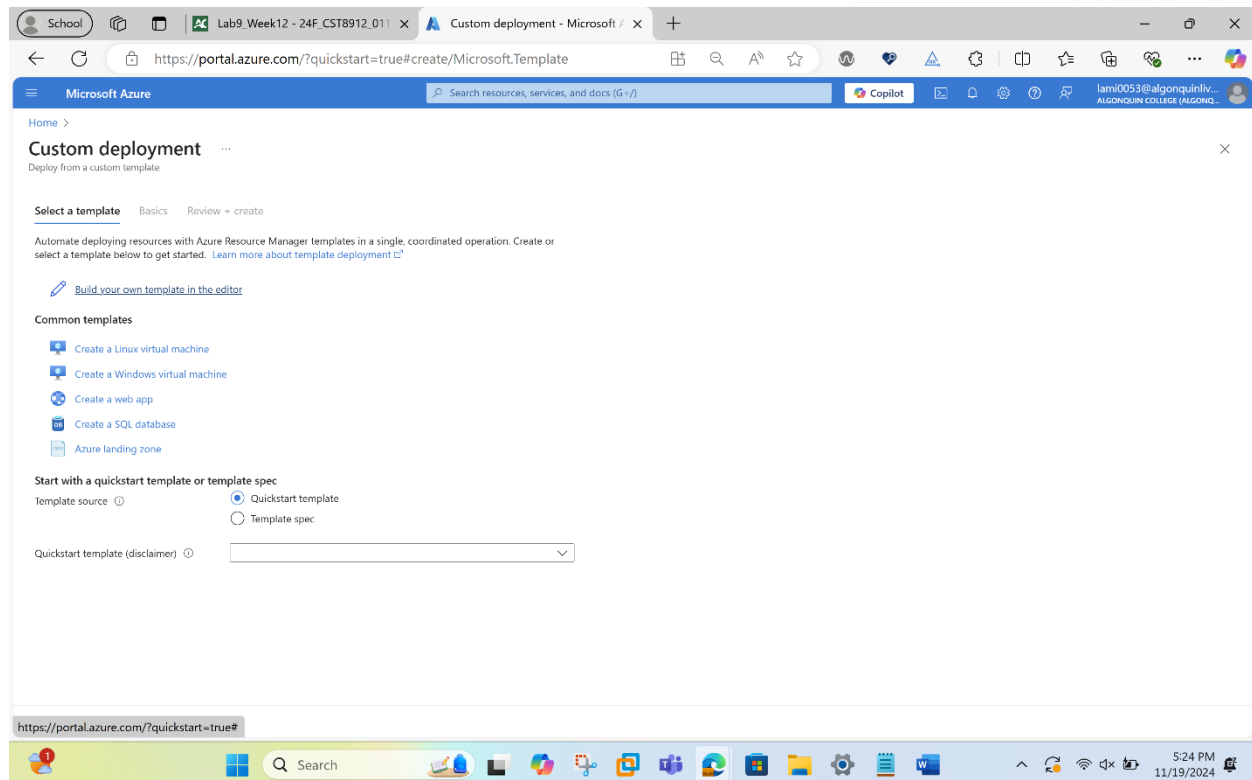
Task 1: Deploy an Azure SQL Database

Method 2:

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, type Deploy a custom template and press the Enter key.



On the **Custom deployment** blade, click the **Build your own template in the editor** option.



On the **Edit template** blade, click **Load file**, locate the <https://github.com/MicrosoftLearning/AZ500-AzureSecurityTechnologies/blob/master/Allfiles/Labs/11/azuredeploy.json> file and click **Open**.

In this step, we changed the location to canadacentral.

School Lab9_Week12 - 24F_CST8912_011 Edit template - Microsoft Azure AZ500-AzureSecurityTechnologies

https://github.com/MicrosoftLearning/AZ500-AzureSecurityTechnologies/blob/m...

MicrosoftLearning / AZ500-AzureSecurityTechnologies

Code Issues Pull requests Actions Security Insights

Files

master

Go to file

.github

Allfiles/Labs

02

03

04

06

07

08

09

10

11

azuredeploy.json

main.bicep

main.bicepparam

12

13

14

15

Instructions

AZ500-AzureSecurityTechnologies / Allfiles / Labs / 11 / azuredeploy.json

polichtm update1 4723c51 · 4 years ago History

Code Blame 58 lines (58 loc) · 2.88 KB

```

1 {
2   "$schema": "https://schema.management.azure.com/schemas/2015-01-01/deploymentTemplate.json#",
3   "contentVersion": "1.0.0.0",
4   "parameters": {
5   },
6   "variables": {
7     "sqlAdministratorLogin": "sqladmin",
8     "sqlAdministratorLoginPassword": "Pa55w.rd",
9     "sqlServerName": "[concat('az50011',uniqueString(resourceGroup().id))]"
10  },
11  "resources": [
12    {
13      "type": "Microsoft.Sql/servers",
14      "apiVersion": "2015-05-01-preview",
15      "name": "[variables('sqlServerName')]",
16      "location": "eastus",
17      "kind": "v12.0",
18      "properties": {
19        "administratorLogin": "[variables('sqlAdministratorLogin')]",
20        "administratorLoginPassword": "[variables('sqlAdministratorLoginPassword')]",
21        "version": "12.0"
22      }
23    },
24    {
25      "type": "Microsoft.Sql/servers/firewallRules",
26      "apiVersion": "2015-05-01-preview",
27      "name": "[concat(variables('sqlServerName'), '/AllowAllWindowsAzureIps')]",
28      "dependsOn": [

```

5:24 PM 11/19/2024

School Lab9_Week12 - 24F_CST8912_011 Edit template - Microsoft Azure AZ500-AzureSecurityTechnologies

https://portal.azure.com/?quickstart=true#view/HubsExtension/TemplateEditorBl...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

lami0053@algonquinlv... ALGONQUIN COLLEGE (ALGONQ...

Home > Custom deployment >

Edit template

Edit your Azure Resource Manager template

+ Add resource ↑ Quickstart template ↑ Load file ↓ Download

Parameters (0)

Variables (3)

Resources (3)

[variables('sqlServerName')]
(Microsoft.Sql/servers)

[concat(variables('sqlServerName'), '/AllowAllWindowsAzureIps')]
(Microsoft.Sql/servers/firewallRule)

[concat(variables('sqlServerName'), '/AZ500LabDb')]
(Microsoft.Sql/servers/databases)

```

17 "kind": "v12.0",
18 "properties": {
19   "administratorLogin": "[variables('sqlAdministratorLogin')]",
20   "administratorLoginPassword": "[variables('sqlAdministratorLoginPassword')]",
21   "version": "12.0"
22 },
23 },
24 {
25   "type": "Microsoft.Sql/servers/firewallRules",
26   "apiVersion": "2015-05-01-preview",
27   "name": "[concat(variables('sqlServerName'), '/AllowAllWindowsAzureIps')]",
28   "dependsOn": [
29     "[resourceId('Microsoft.Sql/servers', variables('sqlServerName'))]"
30   ],
31   "properties": {
32     "startIpAddress": "0.0.0.0",
33     "endIpAddress": "0.0.0.0"
34   }
35 },
36 {
37   "type": "Microsoft.Sql/servers/databases",
38   "apiVersion": "2017-03-01-preview",
39   "name": "[concat(variables('sqlServerName'), '/AZ500LabDb')]",
40   "location": "canadacentral",
41   "dependsOn": [
42     "[resourceId('Microsoft.Sql/servers', variables('sqlServerName'))]"
43   ],
44   "sku": {
45     "name": "Basic",
46     "tier": "Basic"
47   },

```

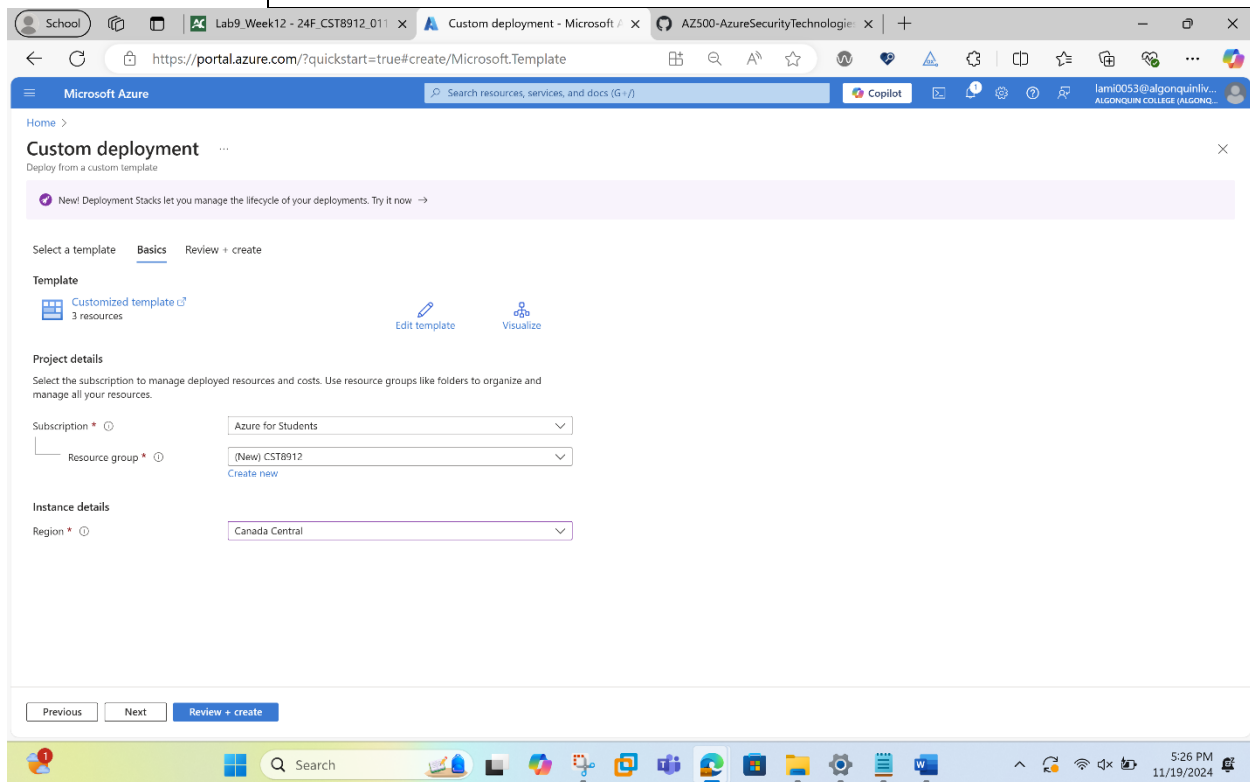
Save Discard

5:39 PM 11/19/2024

On the **Edit template** blade, click **Save**.

On the **Custom deployment** blade, ensure that the following settings are configured (leave any others with their default values):

Setting	Value
Subscription	Azure subscription
Resource group	CST8912
Location	Canada Central



The screenshot shows the 'Custom deployment' blade in the Microsoft Azure portal. The 'Basics' tab is selected, and the 'Project details' section is visible. The configuration is as follows:

- Subscription:** Azure for Students
- Resource group:** (New) CST8912
- Region:** Canada Central

At the bottom of the blade, there are three buttons: 'Previous', 'Next', and 'Review + create'.

Click **Review + Create** and then click **Create**.

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School | Lab9_Week12 - 24F_CST8912_011 | Custom deployment - Microsoft | AZ500-AzureSecurityTechnologie | +

https://portal.azure.com/?quickstart=true#create/Microsoft.Template

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Custom deployment | Deploy from a custom template

Select a template | Basics | Review + create

Summary

- Customized template
- 3 resources

Terms

[Azure Marketplace Terms](#) | [Azure Marketplace](#)

By clicking "Create," I (a) agree to the applicable legal terms associated with the offering; (b) authorize Microsoft to charge or bill my current payment method for the fees associated the offering(s), including applicable taxes, with the same billing frequency as my Azure subscription, until I discontinue use of the offering(s); and (c) agree that, if the deployment involves 3rd party offerings, Microsoft may share my contact information and other details of such deployment with the publisher of that offering.

Microsoft assumes no responsibility for any actions performed by third-party templates and does not provide rights for third-party products or services. See the [Azure Marketplace Terms](#) for additional terms.

Deploying this template will create one or more Azure resources or Marketplace offerings. You acknowledge that you are responsible for reviewing the applicable pricing and legal terms associated with all resources and offerings deployed as part of this template. Prices and associated legal terms for any Marketplace offerings can be found in the [Azure Marketplace](#); both are subject to change at any time prior to deployment.

Neither subscription credits nor monetary commitment funds may be used to purchase non-Microsoft offerings. These purchases are billed separately.

If any Microsoft products are included in a Marketplace offering (e.g. Windows Server or SQL Server), such products are licensed by Microsoft and not by any third party.

Basics

Customization | [Azure for Students](#)

Previous | Next | Create

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School | Lab9_Week12 - 24F_CST8912_011 | Microsoft.Template-20241119174 | AZ500-AzureSecurityTechnologie | +

https://portal.azure.com/?quickstart=true#view/HubsExtension/Deploym...

Microsoft Azure | Search resources, services, and docs (G+/)

Home > Microsoft.Template-20241119174616 | Overview | Deployment

Search | Delete | Cancel | Redeploy | Download | Refresh

Overview | Inputs | Outputs | Template

✓ Your deployment is complete

Deployment name : Microsoft.Template-20241119174616
Subscription : Azure for Students
Resource group : CST8912
Start time : 11/19/2024, 5:46:18 PM
Correlation ID : cf931ff7-8d76-41a1-9672-c2ac11775391

> Deployment details

Next steps

Go to resource group

Give feedback

Tell us about your experience with deployment

Cost management

Get notified to stay within your budget and prevent unexpected charges on your bill.

Set up cost alerts >

Microsoft Defender for Cloud

Secure your aaos and infrastructure

Notifications

More events in the activity log → | Dismiss all

- ✓ Deployment succeeded
Deployment 'Microsoft.Template-20241119174616' to resource group 'CST8912' was successful.
[Pin to dashboard](#) [Go to resource group](#)
a few seconds ago
- ✓ Deleted resource group cst8912-demo
Deleted resource group cst8912-demo
4 minutes ago
- ✓ Upload Completed for azuredeploy.json
2.08 KiB | "Streaming upload"
9 minutes ago

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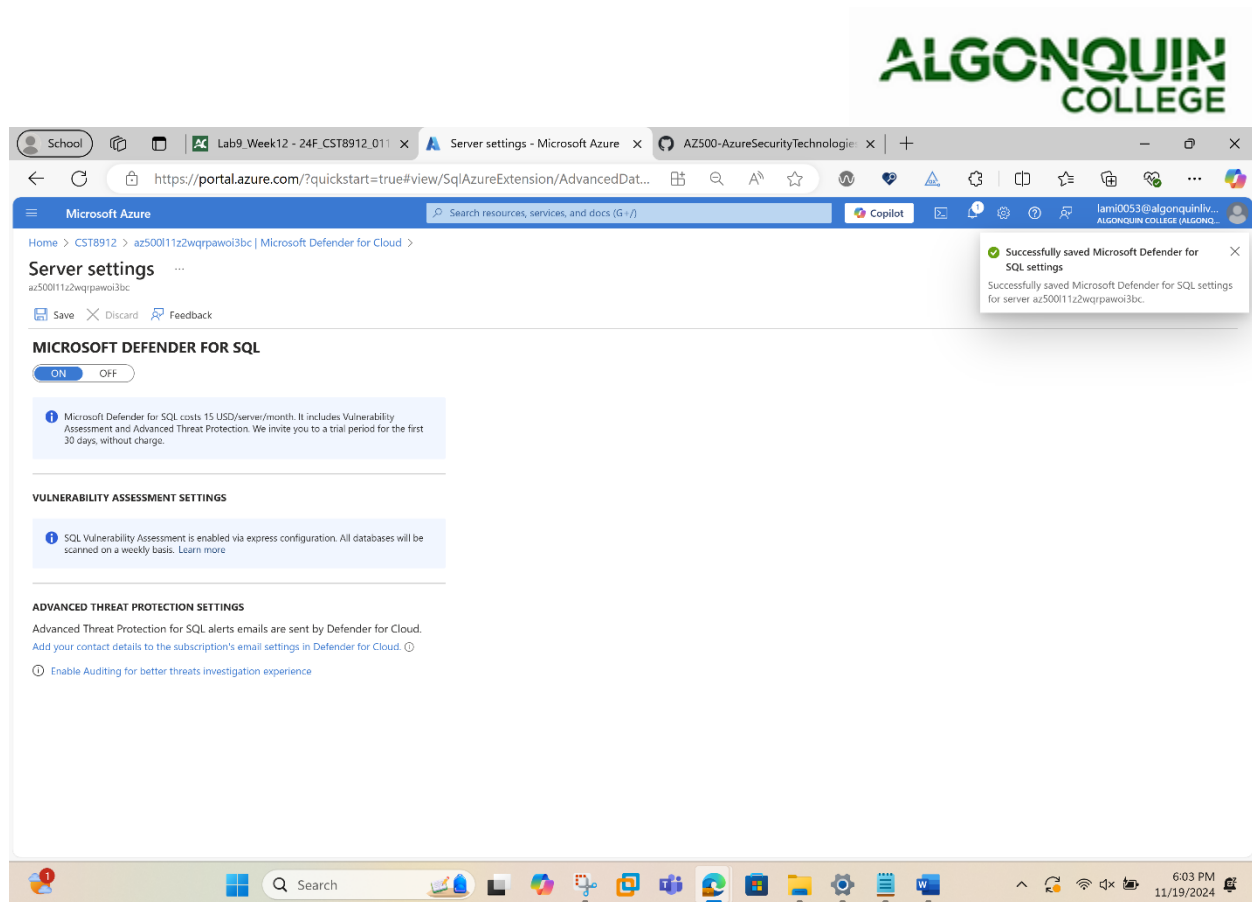
Task 2: Configure Advanced Data Protection

Keshav Lamichhane -Lab9

On the SQL server blade, in the Security section, click Microsoft Defender for Cloud, select Enable Microsoft Defender for SQL.

On the SQL server blade, in the Security section, on the Microsoft Defender for Cloud page, in the Microsoft Defender for SQL: Enabled at the subscription-level (Configure) parameter, click (configure)

The screenshot displays the Microsoft Azure portal interface. The top navigation bar shows the user is logged in as 'lami0053@algonquinliv...'. The main content area is titled 'az50011z2wqrpawoi3bc' and 'SQL server'. The left sidebar contains a navigation menu with options like Overview, Activity log, Access control (IAM), Tags, Quick start, Diagnose and solve problems, Settings, Data management, Security, Networking, Microsoft Defender for Cloud, Transparent data encryption, Identity, Auditing, Intelligent performance, Monitoring, Automation, and Help. The 'Security' section is expanded, showing 'Microsoft Defender for Cloud' and 'Microsoft Defender for SQL'. The 'Microsoft Defender for SQL' tile is highlighted, showing it is 'NOT CONFIGURED'. Below this, there are tiles for 'Automatic tuning' (CONFIGURED), 'Auditing' (NOT CONFIGURED), 'Failover groups' (NOT CONFIGURED), and 'Transparent data encryption' (SERVICE-MANAGED KEY). The bottom of the page shows a table of 'Available resources' with columns for Name, Type, Status, and Pricing tier.



On the Server Settings blade, review the information about pricing and the trial period, VULNERABILITY ASSESSMENT SETTINGS and ADVANCED THREAT PROTECTION SETTINGS.

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School | Lab9_Week12 - 24F-CST8912_011 | Recommendations (Preview) - Microsoft Azure | AZ500-AzureSecurityTechnologies | +

https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Security/Recom...

Microsoft Azure | Search resources, services, and docs (G+)

Home > az5001122wqrpawoi3bc | Microsoft Defender for Cloud >

Recommendations (Preview)

Refresh | Download CSV report | Open query | Governance report | Guides & Feedback | Switch to classic view

We are looking for your feedback! Share with us your thoughts about the new recommendations experience. [Click here to provide feedback.](#)

Scope: Azure subscriptions 1 | AWS accounts 0 | GCP projects 0 | GitHub connectors 0 | AzureDevOps connectors 0 | GitLab connectors 0 | Docker Hub connectors 0 | JFrog connectors 0

Defender CSPM

Recommendations by risk
Prioritized by resource level risk factors and context. [Learn more](#)

Risk based recommendations
0 Critical | 0 High (0) | 0 Medium (0) | 0 Low (0)

Other metrics
0 Active attack paths | 0 Overdue recommendations

Foundational CSPM

Recommendations
0 No risk calculated

Search by title / resource | Status == 3 selected | Risk factors == All | Risk level == All | Recommendation maturity == All | Add filter | Group by title: [dropdown]

Title	Affected resource	Risk level	Risk factors	Attack paths	Owner	Status	Insights
No alerts found							

Give us feedback

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Back to Microsoft Defender for Cloud blade, review Recommendations and Security alerts.

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https://portal.azure.com/?quickstart=true#view/Microsoft_Azure_Security_Azure...

Microsoft Azure | Search resources, services, and docs (G+)

Home > az5001122wqrpawoi3bc | Microsoft Defender for Cloud >

Security alerts

Refresh | Change status | Open query | Suppression rules | Security alerts map | Sample alerts | Alerts workbook | Download CSV report | Guides & Feedback

Open alerts by severity

0 Open alerts | 0 Active alerts | 0 In progress alerts | 0 Affected resources

Search by ID, IP, name, or affected resource... | Subscription == Azure for Students | Status == Active, In Progress | Severity == Low, Medium, High | Affected resource contains az5001122wqrpawoi3bc | Add filter

Severity | Alert name | Affected resource | Resource Group | Activity start time (UTC) | Last updated time (UTC) | MITRE ATT&CK tactics | Status

No alerts found

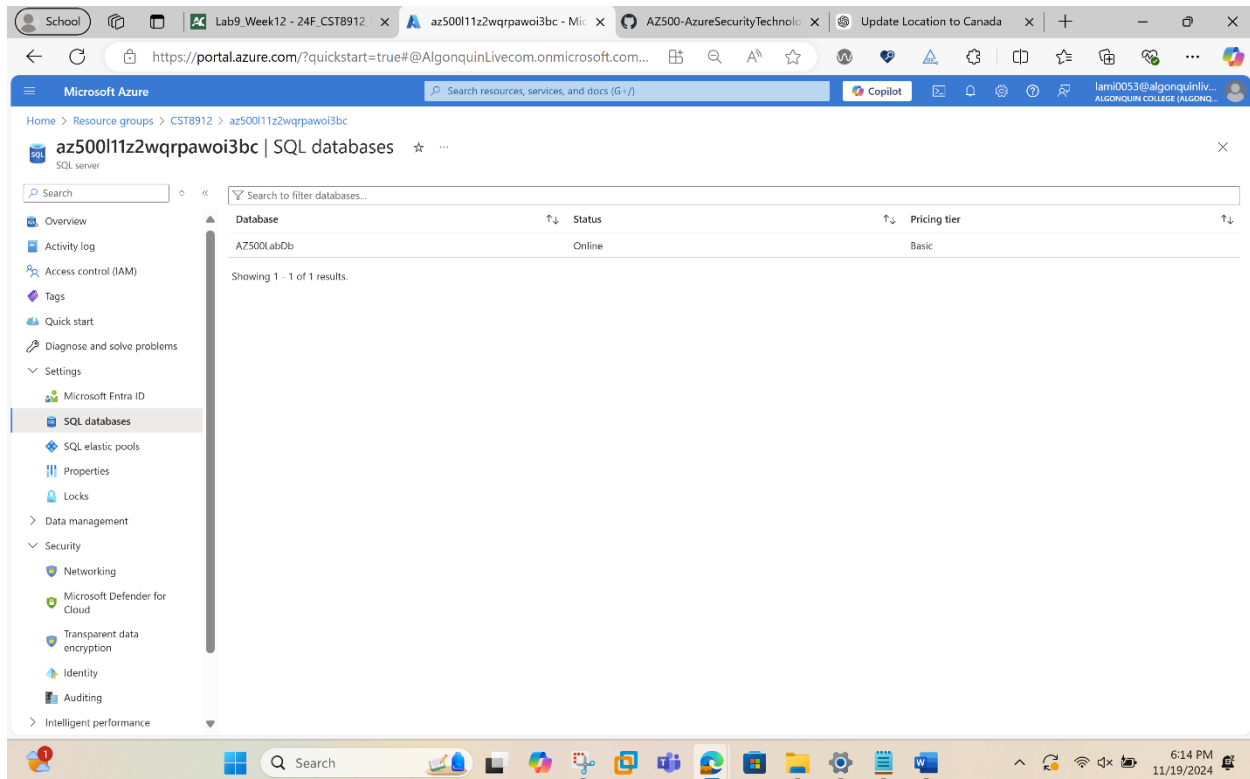
< Previous | Page 1 of 0 | Next >

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Task 3: Configure Data Classification

In this task, you will explore and classify data in SQL database for GPDR and data protection compliance.

On the SQL server blade, in the Settings section, click SQL Databases.



On the SQL database blade, in the Security section, click Data Discovery & Classification.

On the Data Discovery & Classification blade, click the Classification tab.

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School

Lab9_Week12 - 24F-CST8912_011

AZ500LabDb (az50011z2wqrpawoi3bc)

AZ500-AzureSecurityTechnologie

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com...

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

lami0053@algonquinliv... ALGONQUIN COLLEGE (ALGONQ...

Home > az50011z2wqrpawoi3bc > AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb)

AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb) | Data Discovery & Classification

SQL database

Search

Save Discard + Add classification Feedback

Settings

- Compute + storage
- Connection strings
- Properties
- Locks

Data management

- Integrations
- Power Platform

Security

- Auditing
- Ledger
- Data Discovery & Classification
- Dynamic Data Masking
- Microsoft Defender for Cloud
- Identity
- Data Encryption
- Intelligent performance
- Monitoring
- Automation
- Help

Currently database is using SQL Information Protection policy. Found 15 columns with classification recommendations →

Overview Classification

Learn more - Getting Started Guide

No classified columns

Schema: 0 Table: 0 Filter by column name Information type: 0 Sensitivity label: 0

Schema	Table	Column	Information type	Sensitivity label
--------	-------	--------	------------------	-------------------

< Previous Page 1 of 1 Next >

Currently you do not have any classified columns

Add classification

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Click the text message We have found 15 columns with classification recommendations displayed on blue bar at the top of the blade.

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School | Lab9_Week12 - 24F_CST8912_01 | AZ500LabDb (az50011z2wqrpawoi3bc) | AZ500-AzureSecurityTechnologie | +

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com...

Microsoft Azure | Search resources, services, and docs (G+/)

Home > az50011z2wqrpawoi3bc > AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb)

AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb) | Data Discovery & Classification ☆ ...

SQL database

Search

Settings

- Compute + storage
- Connection strings
- Properties
- Locks

Data management

- Integrations
- Power Platform

Security

- Auditing
- Ledger
- Data Discovery & Classification**
- Dynamic Data Masking
- Microsoft Defender for Cloud
- Identity
- Data Encryption
- Intelligent performance
- Monitoring
- Automation
- Help

Save Discard + Add classification Feedback

Currently database is using SQL Information Protection policy. Found 15 columns with classification recommendations →

Overview Classification

No classified columns

Schema: 0 Table: 0 Filter by column name Information type: 0 Sensitivity label: 0

Schema	Table	Column	Information type	Sensitivity label
--------	-------	--------	------------------	-------------------

< Previous Page 1 of 1 Next >

Currently you do not have any classified columns

Add classification

6:18 PM 11/19/2024

Review the listed columns and the recommended sensitivity label.

Enable the Select all checkbox and then click Accept Selected Recommendations.

School Lab9_Week12 - 24F_CST8912_011 AZ500LabDb (az50011z2wqrpawoi3bc) AZ500-AzureSecurityTechnologie

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com...

Microsoft Azure Search resources, services, and docs (G+/)

Home > az50011z2wqrpawoi3bc > AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb)

AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb) | Data Discovery & Classification

SQL database

Save Discard + Add classification Feedback

Currently database is using SQL Information Protection policy. Found 15 columns with classification recommendations →

Overview Classification [Learn more - Getting Started Guide](#)

15 columns with classification recommendations (Click to minimize)

Accept selected recommendations Dismiss selected recommendations Show dismissed recommendations

Select all Schema: 2 Table: 5 Filter by column name Information type: 5 Sensitivity label: 2

	Schema	Table	Column	Information type	Sensitivity label
☐	SalesLT	Customer	FirstName	Name	Confidential - GDPR
☐	SalesLT	Customer	LastName	Name	Confidential - GDPR
☐	SalesLT	Customer	EmailAddress	Contact Info	Confidential
☐	SalesLT	Customer	Phone	Contact Info	Confidential
☐	SalesLT	Customer	PasswordHash	Credentials	Confidential
☐	SalesLT	Customer	PasswordSalt	Credentials	Confidential
☐	dbo	ErrorLog	UserName	Credentials	Confidential
☐	SalesLT	Address	AddressLine1	Contact Info	Confidential
☐	SalesLT	Address	AddressLine2	Contact Info	Confidential
☐	SalesLT	Address	City	Contact Info	Confidential
☐	SalesLT	Address	PostalCode	Contact Info	Confidential

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School Lab9_Week12 - 24F_CST8912_011 AZ500LabDb (az50011z2wqrpawoi3bc) AZ500-AzureSecurityTechnologie

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com...

Microsoft Azure Search resources, services, and docs (G+/)

Home > az50011z2wqrpawoi3bc > AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb)

AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb) | Data Discovery & Classification

SQL database

Save Discard + Add classification Feedback

Currently database is using SQL Information Protection policy.

Overview Classification [Learn more - Getting Started Guide](#)

15 classified columns

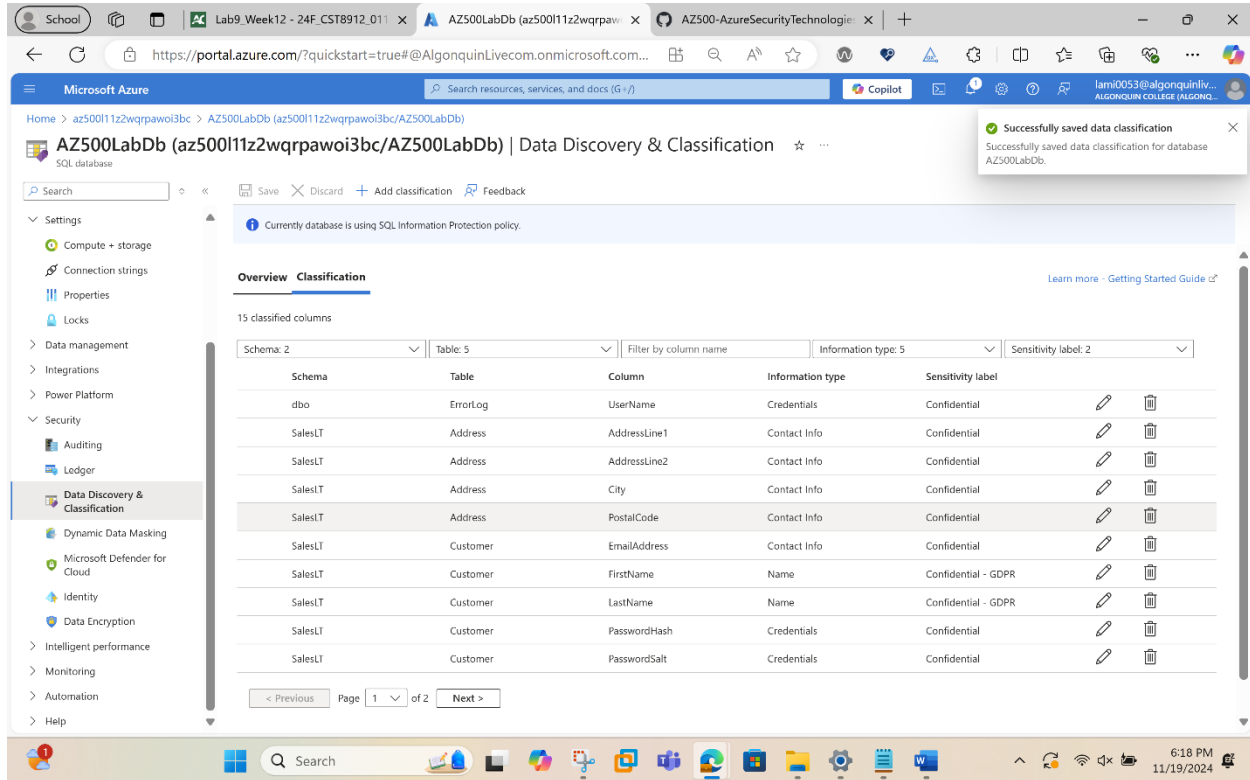
Schema: 2 Table: 5 Filter by column name Information type: 5 Sensitivity label: 2

	Schema	Table	Column	Information type	Sensitivity label		
	dbo	ErrorLog	UserName	Credentials	Confidential		
	SalesLT	Address	AddressLine1	Contact Info	Confidential		
	SalesLT	Address	AddressLine2	Contact Info	Confidential		
	SalesLT	Address	City	Contact Info	Confidential		
	SalesLT	Address	PostalCode	Contact Info	Confidential		
	SalesLT	Customer	EmailAddress	Contact Info	Confidential		
	SalesLT	Customer	FirstName	Name	Confidential - GDPR		
	SalesLT	Customer	LastName	Name	Confidential - GDPR		
	SalesLT	Customer	PasswordHash	Credentials	Confidential		
	SalesLT	Customer	PasswordSalt	Credentials	Confidential		

< Previous Page 1 of 2 Next >

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Once you have completed your review click Save.



Successfully saved data classification
Successfully saved data classification for database AZ500LabDb.

Currently database is using SQL Information Protection policy.

Overview Classification

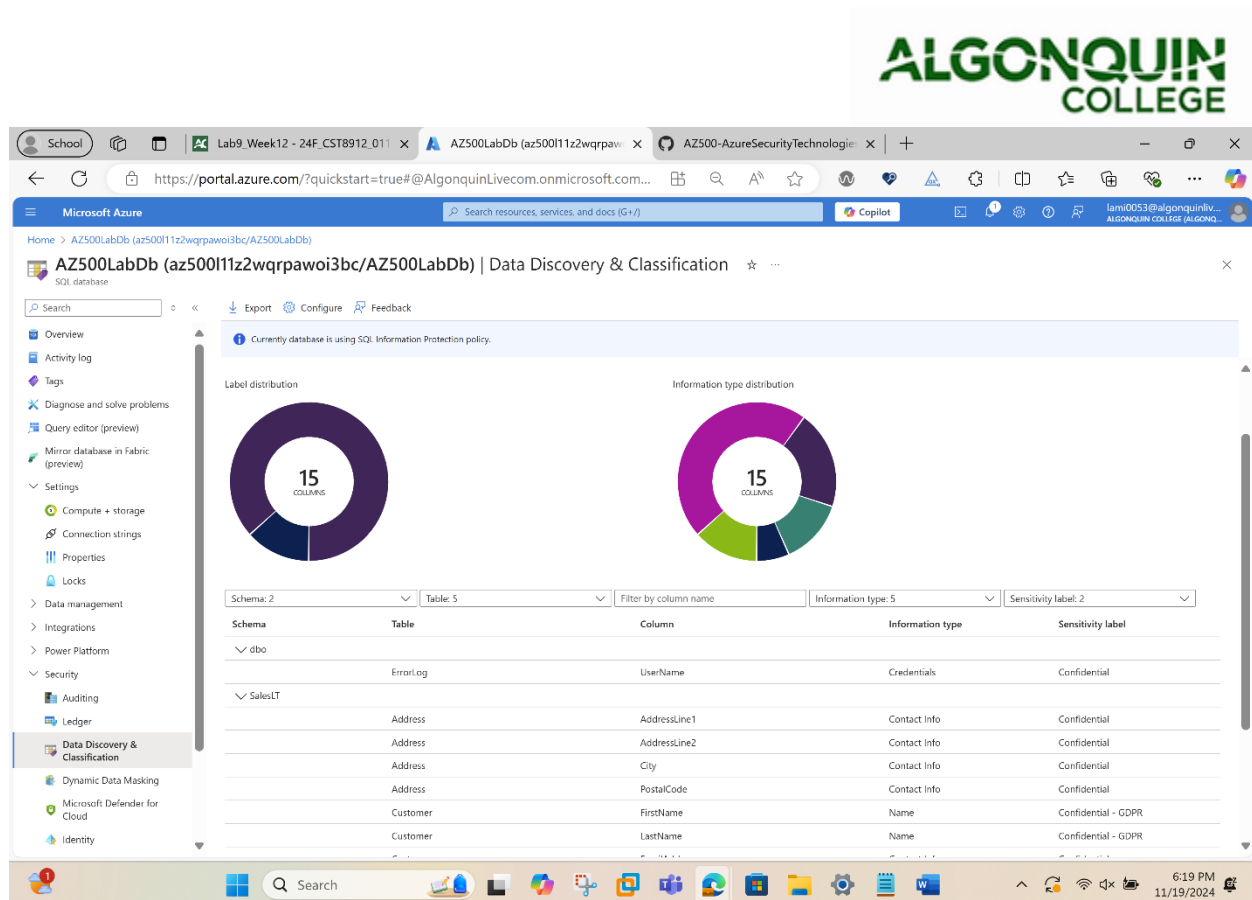
15 classified columns

Schema: 2 Table: 5 Filter by column name Information type: 5 Sensitivity label: 2

Schema	Table	Column	Information type	Sensitivity label
dbo	ErrorLog	UserName	Credentials	Confidential
SalesLT	Address	AddressLine1	Contact Info	Confidential
SalesLT	Address	AddressLine2	Contact Info	Confidential
SalesLT	Address	City	Contact Info	Confidential
SalesLT	Address	PostalCode	Contact Info	Confidential
SalesLT	Customer	EmailAddress	Contact Info	Confidential
SalesLT	Customer	FirstName	Name	Confidential - GDPR
SalesLT	Customer	LastName	Name	Confidential - GDPR
SalesLT	Customer	PasswordHash	Credentials	Confidential
SalesLT	Customer	PasswordSalt	Credentials	Confidential

< Previous Page 1 of 2 Next >

Back on the Data Discovery & Classification blade Overview tab, note that it has been updated to account for the latest classification information.



Task 4: Configure Auditing

In this task, you will first configure server level auditing and then configure database level auditing.

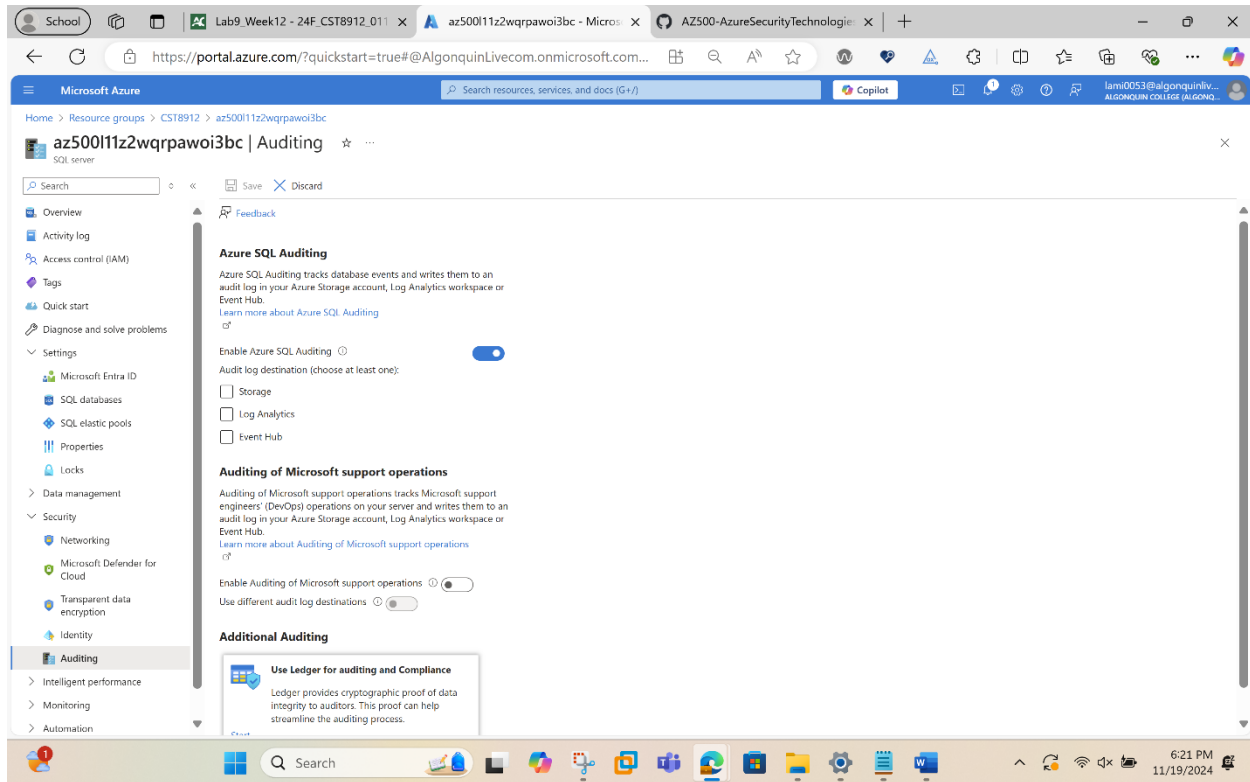
In the Azure portal, navigate back to the SQL Server blade.

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the 'School' tab, a search bar, and the user profile 'lami0053@algonquiniv...'. The main content area displays the 'Overview' page for the SQL server resource 'az50011z2wqrpawoi3bc'. The left sidebar contains a navigation menu with categories like 'Overview', 'Settings', 'Data management', 'Security', and 'Monitoring'. The 'Security' section is expanded, showing 'Auditing' as the selected option. The main content area includes a 'Notifications' section with a 'Microsoft Defender for SQL Free Trial' alert, a 'Features' section, and an 'Available resources' table. The table lists the 'SQL database' resource 'AZ500Lab0b' with a status of 'Online' and a pricing tier of 'Basic'.

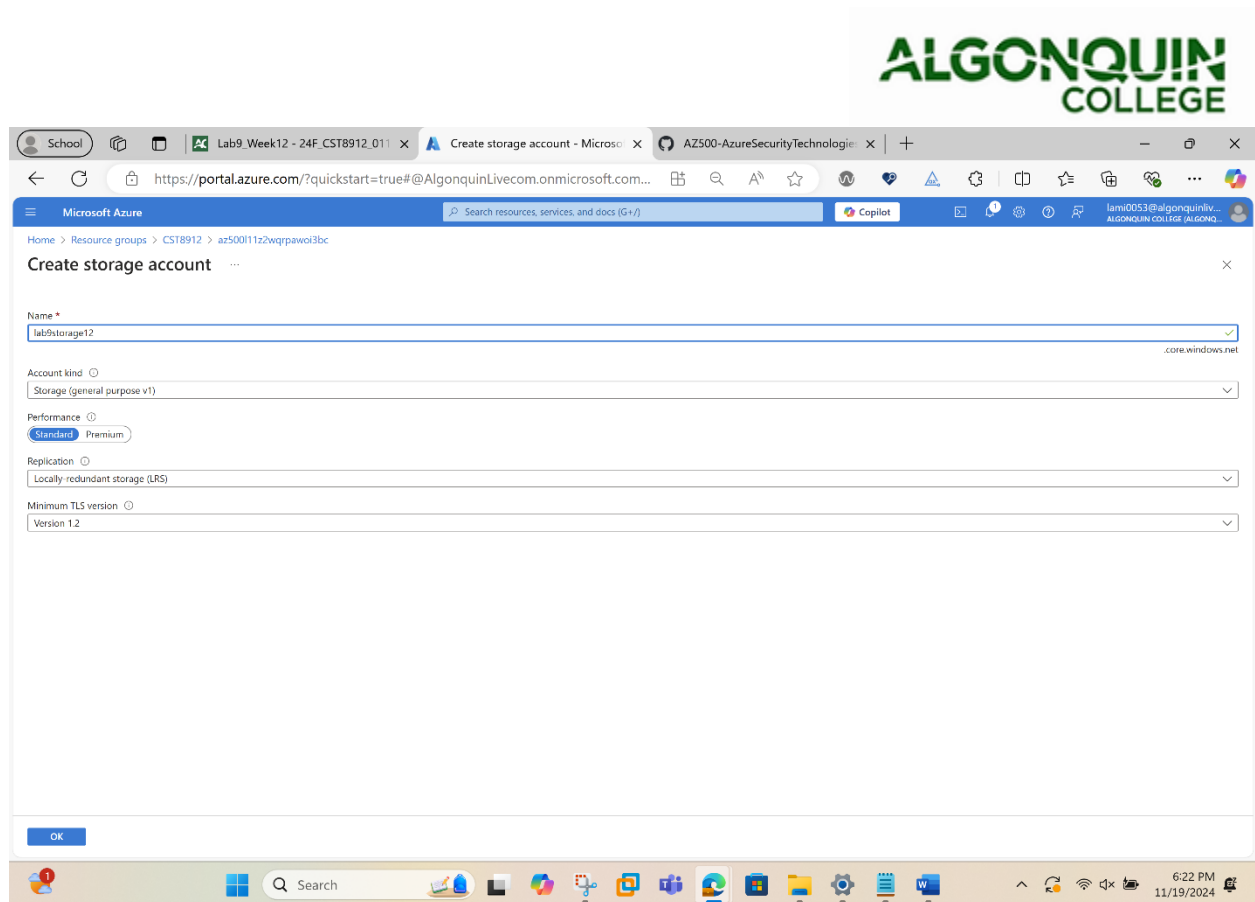
On the SQL Server blade, in the Security section, click Auditing.

The screenshot shows the Microsoft Azure portal interface, specifically the 'Auditing' page for the SQL server resource 'az50011z2wqrpawoi3bc'. The left sidebar navigation menu is the same as in the previous screenshot, but the 'Auditing' option under the 'Security' section is now selected. The main content area displays the 'Azure SQL Auditing' page, which includes a description of the auditing feature, a toggle switch to 'Enable Azure SQL Auditing', and a section for 'Auditing of Microsoft support operations'. There is also a 'Use Ledger for auditing and Compliance' section with a 'Start' button. The bottom of the page shows the Windows taskbar with the search bar and various application icons.

Set the Enable Azure SQL Auditing switch to ON to enable auditing

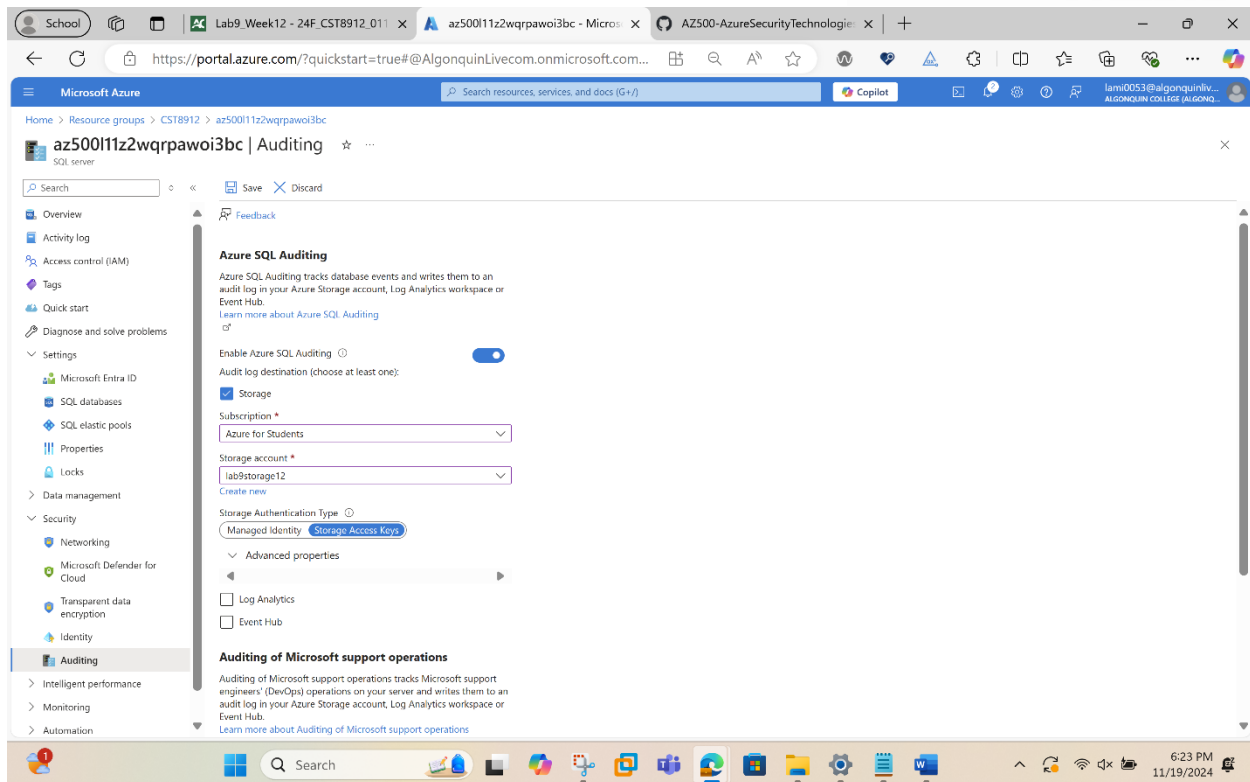


Select the Storage checkbox and entry boxes for Subscription and Storage Account will display (create new storage account if not selected)

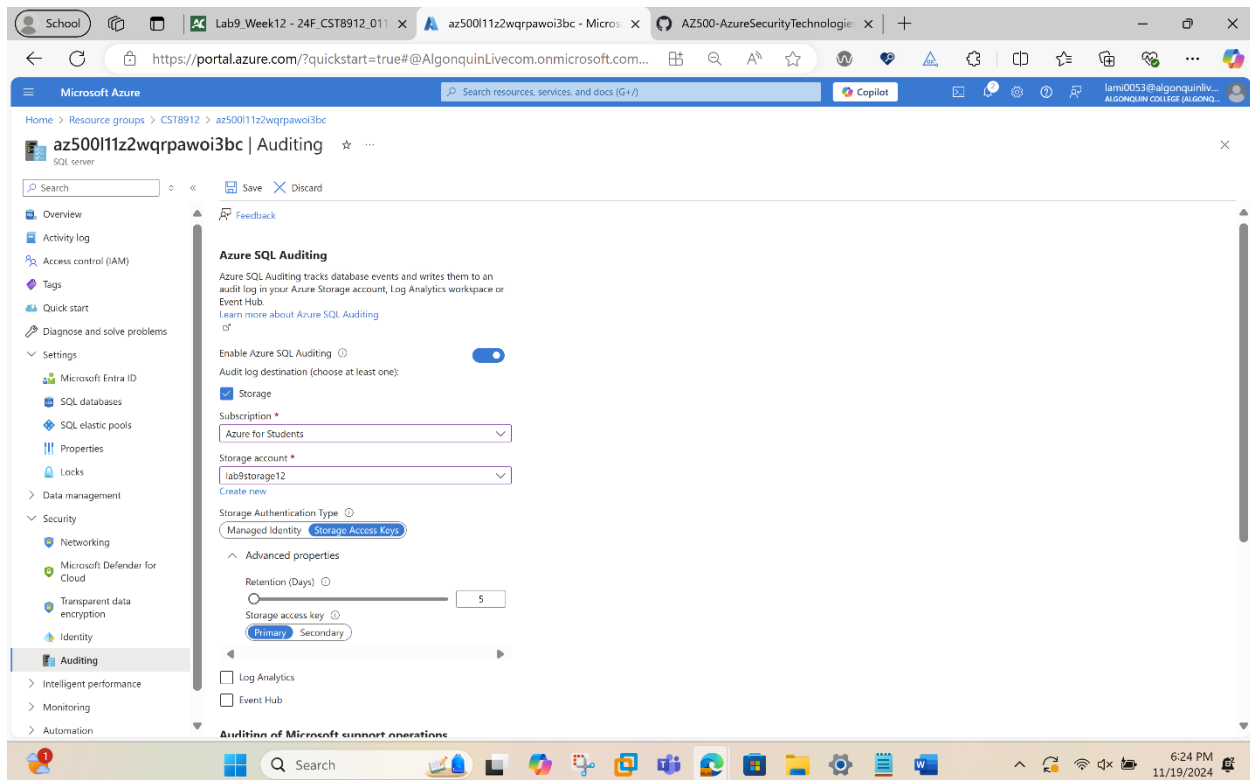


Choose your Subscription from the dropdown list.

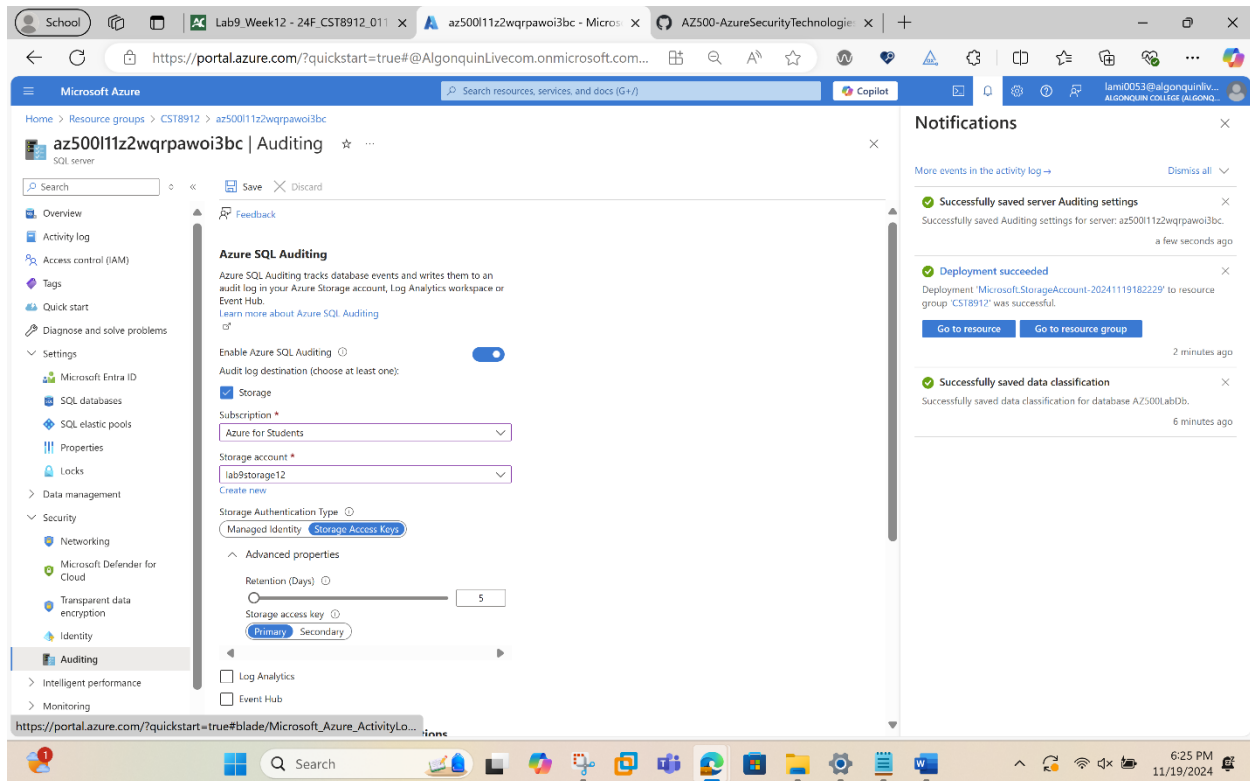
Click Storage account and choose Create new.



Back on the Auditing blade, under Advanced properties set Retention (days) to 5.



On the Auditing blade, click Save to save the auditing settings.



On the server blade, in the Settings section, click SQL Databases.

On the SQL database blade, in the Security section, click Auditing.

Note: This is database level auditing. Server-level auditing is already enabled.

School Lab9_Week12 - 24F: CST8912_011 AZ500LabDb (az50011z2wqrpawoi3bc) AZ500-AzureSecurityTechnologie

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com...

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource groups > CST8912 > az50011z2wqrpawoi3bc | SQL databases > AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb)

AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb) | Auditing

SQL database

Search Save Discard View audit logs

Overview Activity log Tags Diagnose and solve problems Query editor (preview) Mirror database in Fabric (preview) Settings Compute + storage Connection strings Properties Locks Data management Integrations Power Platform Security Auditing Ledger Data Discovery & Classification Dynamic Data Masking Microsoft Defender for Cloud Identity

Feedback

If Blob Auditing is enabled on the server, it will always apply to the database, regardless of the database settings.

View server settings

Server-level Auditing: **Enabled**

Azure SQL Auditing

Azure SQL Auditing tracks database events and writes them to an audit log in your Azure Storage account, Log Analytics workspace or Event Hub.

Learn more about Azure SQL Auditing

Enable Azure SQL Auditing

Additional Auditing

Use Ledger for auditing and Compliance

Ledger provides cryptographic proof of data integrity to auditors. This proof can help streamline the auditing process.

Start

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School Lab9_Week12 - 24F: CST8912 AZ500LabDb (az50011z2wqrpawoi3bc) AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb) AZ500-AzureSecurityTechnologie ChatGPT

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft.com...

Microsoft Azure Search resources, services, and docs (G+)

Home > Resource groups > CST8912 > AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb)

AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb) | Auditing

SQL database

Search Save Discard View audit logs

Overview Activity log Tags Diagnose and solve problems Query editor (preview) Mirror database in Fabric (preview) Settings Compute + storage Connection strings Properties Locks Data management Integrations Azure Synapse Link Stream analytics (preview) Add Azure AI Search Power Platform Security Auditing Ledger Data Discovery & Classification

Feedback

If Blob Auditing is enabled on the server, it will always apply to the database, regardless of the database settings.

View server settings

Server-level Auditing: **Enabled**

Azure SQL Auditing

Azure SQL Auditing tracks database events and writes them to an audit log in your Azure Storage account, Log Analytics workspace or Event Hub.

Learn more about Azure SQL Auditing

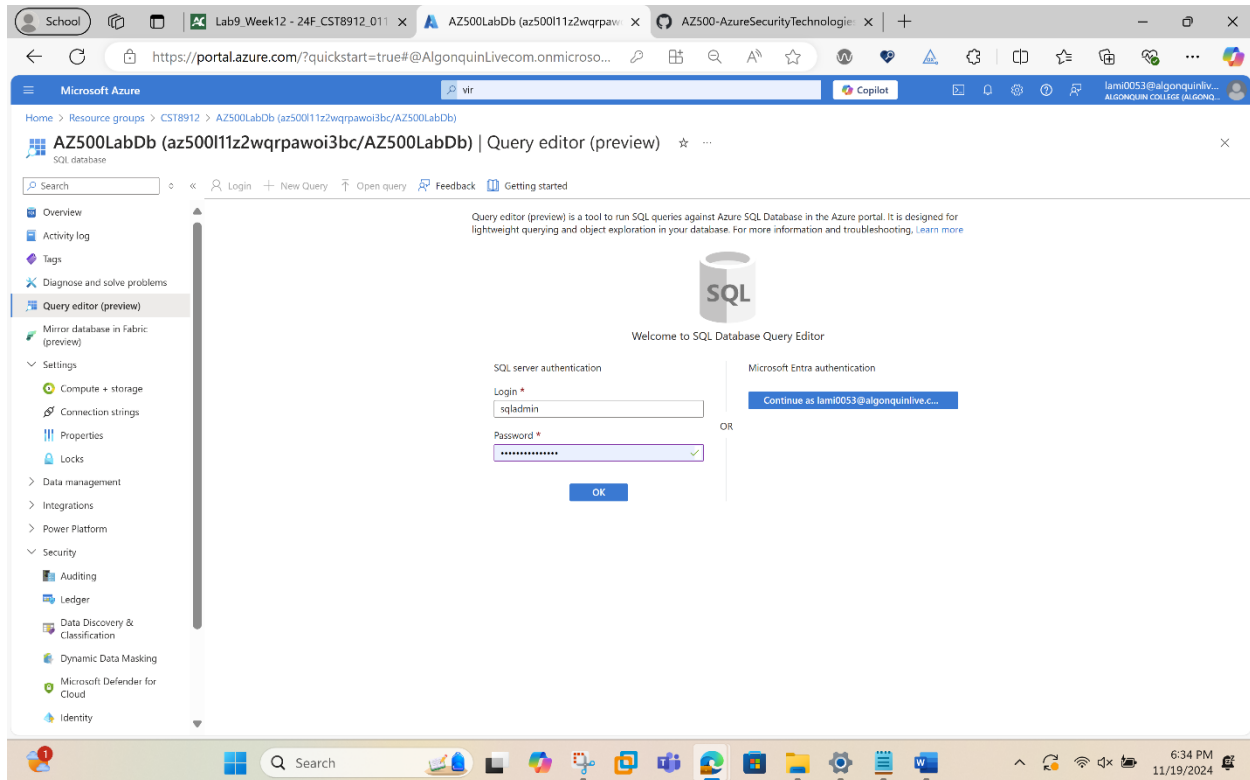
Enable Azure SQL Auditing

Audit log destination (choose at least one):

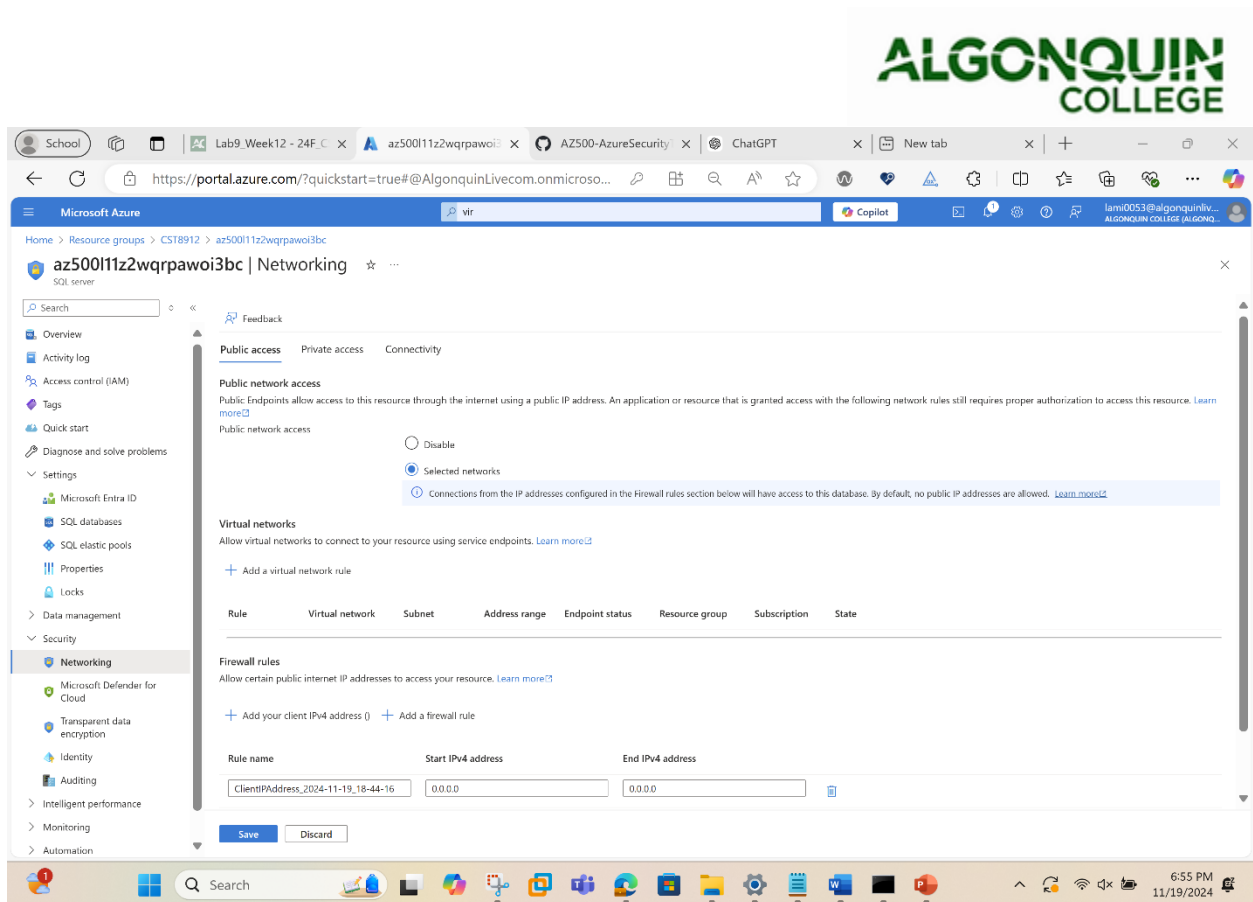
- ☒ Storage
 - Subscription: Azure for Students
 - Storage account: lab9storage12
 - Create new
 - Storage Authentication Type: Managed Identity (Storage Access Keys)
 - Advanced properties
- ☐ Log Analytics
- ☐ Event Hub

7:27 PM 11/19/2024

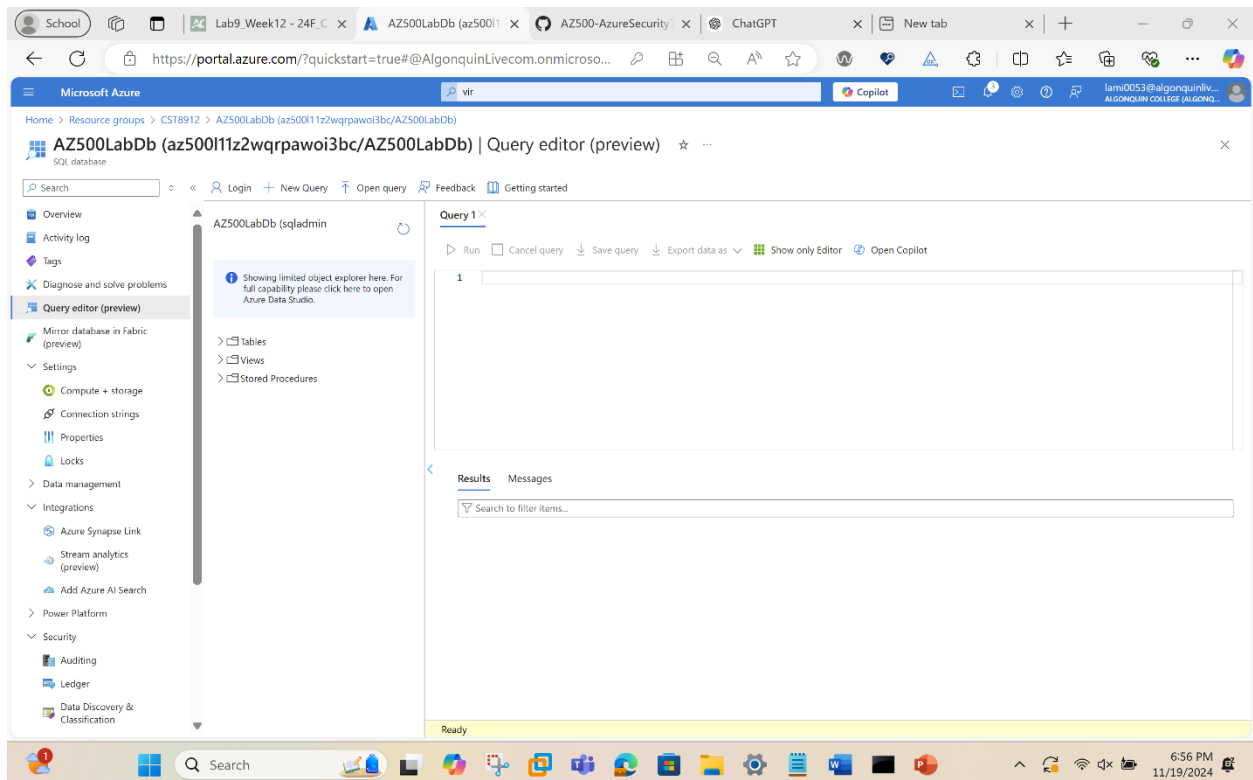
On your SQL database Overview page in the Azure portal, select Query editor (preview) from the left menu. Try to sign in, you might fail on password, firewall rule for your IP address, everything gets audited. Try successful login as well, run query and you might find more details in audit logs.



We then assigned ip address.



We could then login to query editor.



Keshav Lamichhane -Lab9

School Lab9_Week12 - 24F_CST8912 AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb) AZ500-AzureSecurityTechnolo New tab

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicroso...

Microsoft Azure vir Copilot lam0053@algonquinliv... ALGONQUIN COLLEGE (ALGONQ...

Home > Resource groups > CST8912 > AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb)

AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb) | Query editor (preview)

SQL database

Search Login + New Query Open query Feedback Getting started

Overview Activity log Tags Diagnose and solve problems Query editor (preview)

Mirror database in Fabric (preview)

Settings

- Compute + storage
- Connection strings
- Properties
- Locks
- Data management
- Integrations
 - Azure Synapse Link
 - Stream analytics (preview)
 - Add Azure AI Search
- Power Platform
- Security
 - Auditing
 - Ledger
 - Data Discovery & Classification

AZ500LabDb (sqladmin)

Showing limited object explorer here. For full capability please click here to open Azure Data Studio.

Tables

- dbo.BuildVersion
- dbo.ErrorLog
- SalesLT.Address
- SalesLT.Customer
- SalesLT.CustomerAddress
- SalesLT.Product
- SalesLT.ProductCategory
- SalesLT.ProductDescription
- SalesLT.ProductModel
- SalesLT.ProductModelProductDesc
- SalesLT.SalesOrderDetail
- SalesLT.SalesOrderHeader
- Stored Procedures

Query 1

Run Cancel query Save query Export data as Show only Editor Open Copilot

```
1 select * from [SalesLT].[Customer]
```

Results Messages

Search to filter items...

CustomerID	NameStyle	Title	FirstName	MiddleName	LastName
1	False	Mr.	Orlando	N.	Gee
2	False	Mr.	Keith		Harris
3	False	Ms.	Donna	F.	Carreras
4	False	Ms.	Janet	M.	Gates
5	False	Mr.	Lucy		Harrington
6	False	Ms.	Rosmarie	J.	Carroll
7	False	Mr.	Dominic	P.	Gash

Query succeeded | 0s

6:58 PM 11/19/2024

School Lab9_Week12 - 24F_CST8912 Audit records - Microsoft Azure AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb) AZ500-AzureSecurityTechnolo

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Microsoft Azure vir Copilot lam0053@algonquinliv... ALGONQUIN COLLEGE (ALGONQ...

Home > AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb)

AZ500LabDb (az50011z2wqrpawoi3bc/AZ500LabDb) | Query editor (preview)

SQL database

Search Login + New Query Open query Feedback Getting started

Overview Activity log Tags Diagnose and solve problems Query editor (preview)

Mirror database in Fabric (preview)

Settings

- Compute + storage
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AZ500LabDb (sqladmin)

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- SalesLT.ProductModelProductDesc
- SalesLT.SalesOrderDetail
- SalesLT.SalesOrderHeader
- Stored Procedures

Query 1

Run Cancel query Save query Export data as Show only Editor Open Copilot

```
1 UPDATE [SalesLT].[Customer]
2 SET [MiddleName] = 'J'
3 WHERE [FirstName] = 'Orlando' AND [LastName] = 'Gee';
4
```

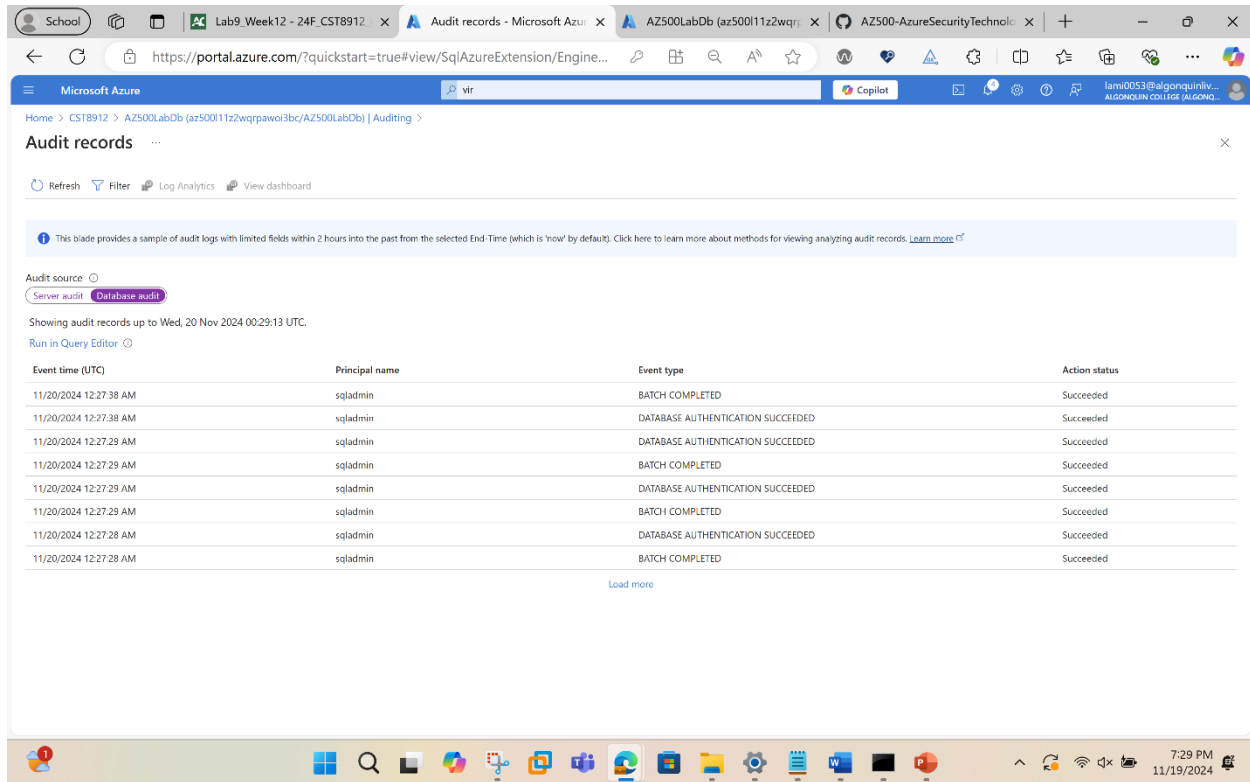
Results Messages

Query succeeded: Affected rows: 2

7:29 PM 11/19/2024

switch back to DB, Auditing and Click View Audit Logs.

On the Audit records blade, note that you can switch between Server audit and Database audit.



Microsoft Azure

Home > CST8912 > AZ500LabDb (az500112zwpawoi3bc/AZ500LabDb) | Auditing >

Audit records

Refresh Filter Log Analytics View dashboard

This blade provides a sample of audit logs with limited fields within 2 hours into the past from the selected End Time (which is 'now' by default). Click here to learn more about methods for viewing analyzing audit records. [Learn more](#)

Audit source

Server audit Database audit

Showing audit records up to Wed, 20 Nov 2024 00:29:13 UTC.

[Run in Query Editor](#)

Event time (UTC)	Principal name	Event type	Action status
11/20/2024 12:27:38 AM	sqladmin	BATCH COMPLETED	Succeeded
11/20/2024 12:27:38 AM	sqladmin	DATABASE AUTHENTICATION SUCCEEDED	Succeeded
11/20/2024 12:27:29 AM	sqladmin	DATABASE AUTHENTICATION SUCCEEDED	Succeeded
11/20/2024 12:27:29 AM	sqladmin	BATCH COMPLETED	Succeeded
11/20/2024 12:27:29 AM	sqladmin	DATABASE AUTHENTICATION SUCCEEDED	Succeeded
11/20/2024 12:27:29 AM	sqladmin	BATCH COMPLETED	Succeeded
11/20/2024 12:27:28 AM	sqladmin	DATABASE AUTHENTICATION SUCCEEDED	Succeeded
11/20/2024 12:27:28 AM	sqladmin	BATCH COMPLETED	Succeeded

[Load more](#)

School Lab9 Week12 - 24F.CST8912 Audit records - Microsoft Azure AZ500LabDb (az500112wqrpawoi3bc/AZ500LabDb) AZ500-AzureSecurityTechno...

https://portal.azure.com/?quickstart=true#view/SqlAzureExtension/Engine...

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Home > CST8912 > AZ500LabDb (az500112wqrpawoi3bc/AZ500LabDb) | Auditing >

Audit records

Refresh Filter Log Analytics View dashboard

This blade provides a sample of audit logs with limited fields within 2 hours into the past from the selected End-Time (which is 'now' by default). Click here to learn more about methods for viewing analyzing audit records. [Learn more](#)

Audit source
[Server audit](#) [Database audit](#)

Showing audit records up to Wed, 20 Nov 2024 00:29:13 UTC.

[Run in Query Editor](#)

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11/20/2024 12:27:29 AM	sqladmin	BATCH COMPLETED	Succeeded
11/20/2024 12:27:29 AM	sqladmin	DATABASE AUTHENTICATION SUCCEEDED	Succeeded
11/20/2024 12:27:29 AM	sqladmin	BATCH COMPLETED	Succeeded
11/20/2024 12:27:28 AM	sqladmin	DATABASE AUTHENTICATION SUCCEEDED	Succeeded
11/20/2024 12:27:28 AM	sqladmin	BATCH COMPLETED	Succeeded
11/20/2024 12:24:28 AM	sqladmin	BATCH COMPLETED	Succeeded
11/20/2024 12:24:28 AM	sqladmin	DATABASE AUTHENTICATION SUCCEEDED	Succeeded

[Load more](#)

Task 5: Clean up resources created during this lab

School Lab9 Week12 Delete a resource group Home - Microsoft Azure AZ500LabDb (az500112wqrpawoi3bc/AZ500LabDb) Facebook New tab

https://portal.azure.com/?quickstart=true#@AlgonquinLivecom.onmicrosoft...

Microsoft Azure vir Copilot lam0053@algonquiniv... ALGONQUIN COLLEGE (ALGONQ...

Home > Resource groups >

Resource groups

Algonquin College (AlgonquinLivecom.onmicrosoft...)

+ Create ... Group by none

You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.

☐ Name 1

☒ CST8912

☐ ResourceMoverRG-eastus-canada

CST8912 Resource group

Search

+ Create Manage view Delete resource group Refresh

Essentials

Subscription (moved) : [Azure for Students](#)

Subscription ID : 469222a6-9aad-44fa-b0b8-120524695b23

Tags (edit) : [Add tags](#)

Resources Recommendations

Filter for any field... type equals all Location equals all

Showing 1 to 3 of 3 records. ☐ Show hidden types

☐ Name ↑

- ☒ az500112wqrpawoi3bc
- ☒ Az500LabDb (az500112wqrpawoi3bc/AZ500LabDb)
- ☒ lab9storage12

Delete a resource group

The following resource group and all its dependent resources will be permanently deleted.

Resource group to be deleted

☒ CST8912

Dependent resources to be deleted (4)

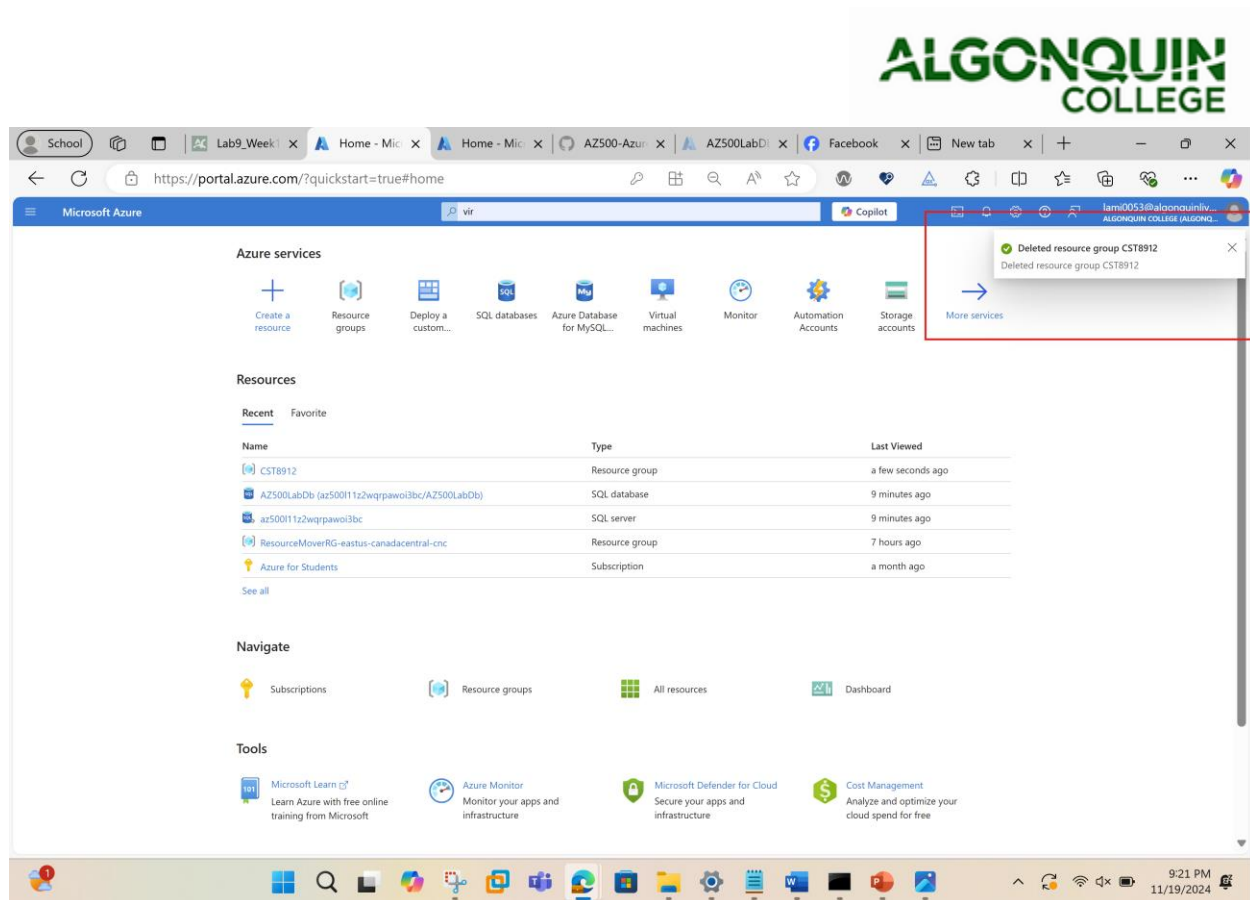
All dependent resources, including hidden types, are shown

Name	Resource type
az500112wqrpawoi3bc	SQL server
AZ500LabDb (az500112wqrpawoi3bc/AZ500	SQL database
lab9storage12	Storage account
master (az500112wqrpawoi3bc/master)	SQL database

Enter resource group name to confirm deletion *

CST8912

Delete Cancel



This step then concludes the lab.

Conclusion: This lab provides hands-on experience with key security features in Azure SQL Database, including advanced data protection, data classification, and auditing. These practices are essential for maintaining the privacy and integrity of client data, ensuring compliance, and enhancing the overall security posture of cloud-based database solutions.